ASSIGNMENT-3

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BATCH-16

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Zero-Shot Prompting – Palindrome Number Program:-

Thonny - <untitled> @ 10 : 1

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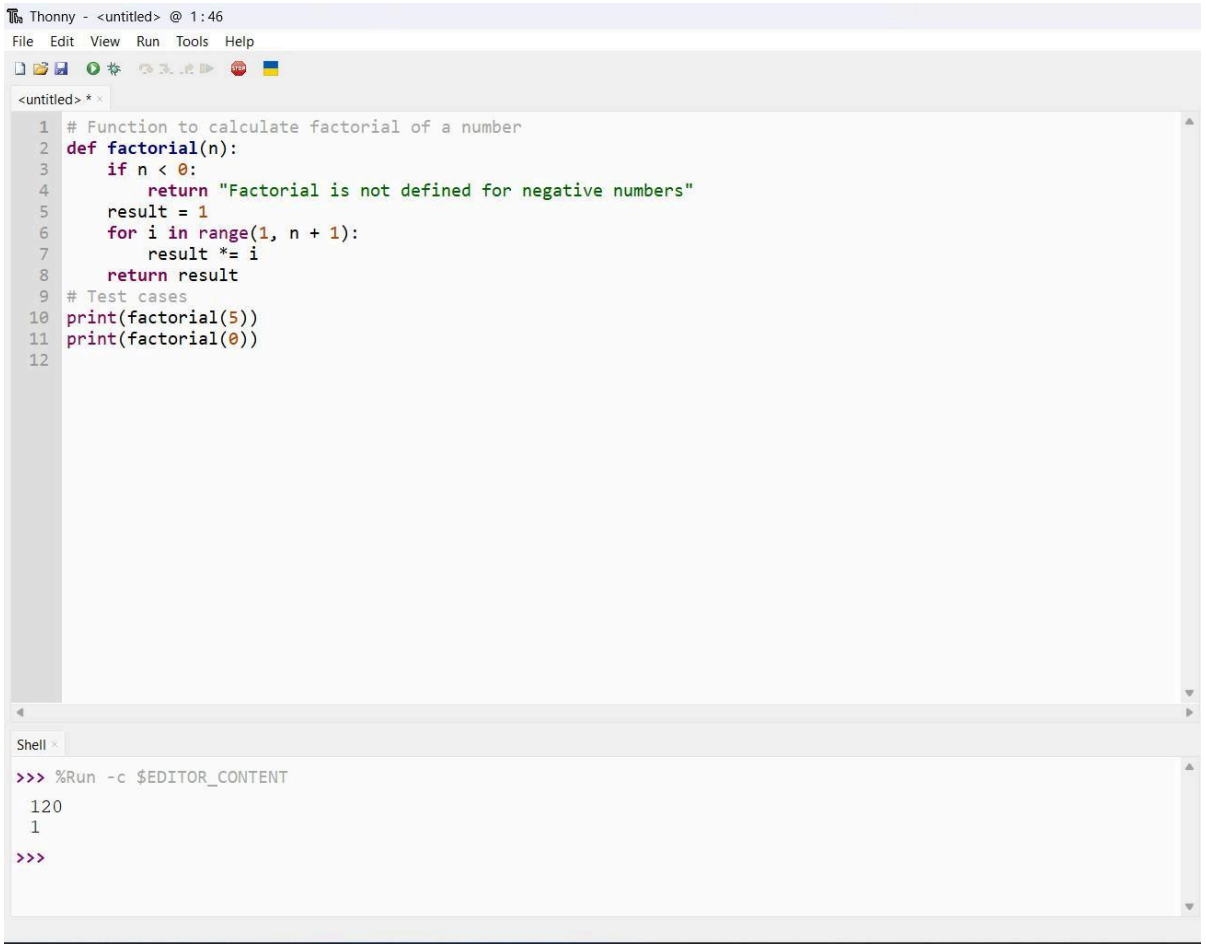
```python
# Function to check whether a number is a palindrome
def is_palindrome(num):
    original = num
    reverse = 0
    while num > 0:
        digit = num % 10
        reverse = reverse * 10 + digit
        num //= 10
    return original == reverse
# Test cases
print(is_palindrome(121))
print(is_palindrome(123))
print(is_palindrome(0))
```

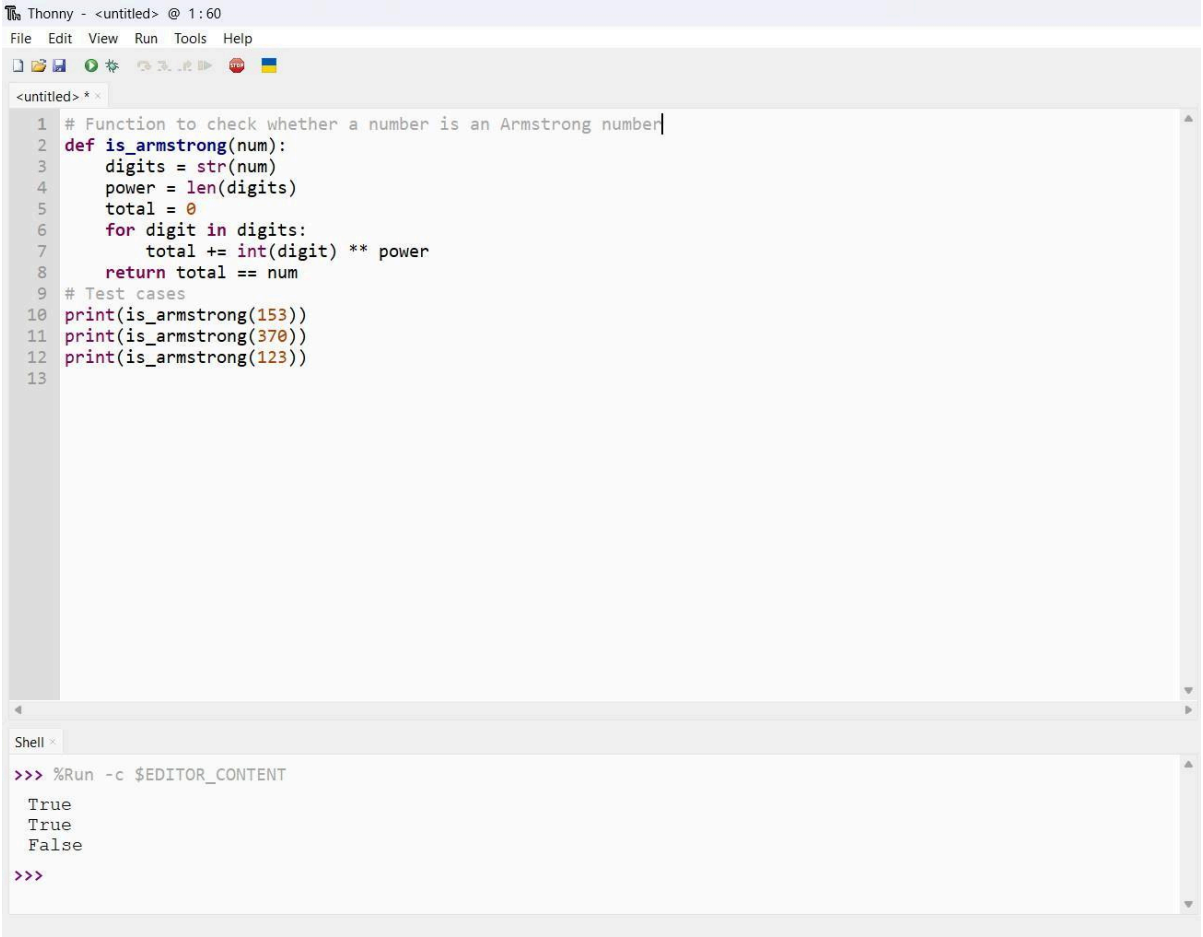
Shell

```
>>> %Run -c $EDITOR_CONTENT
 True
 False
 True
>>>
```

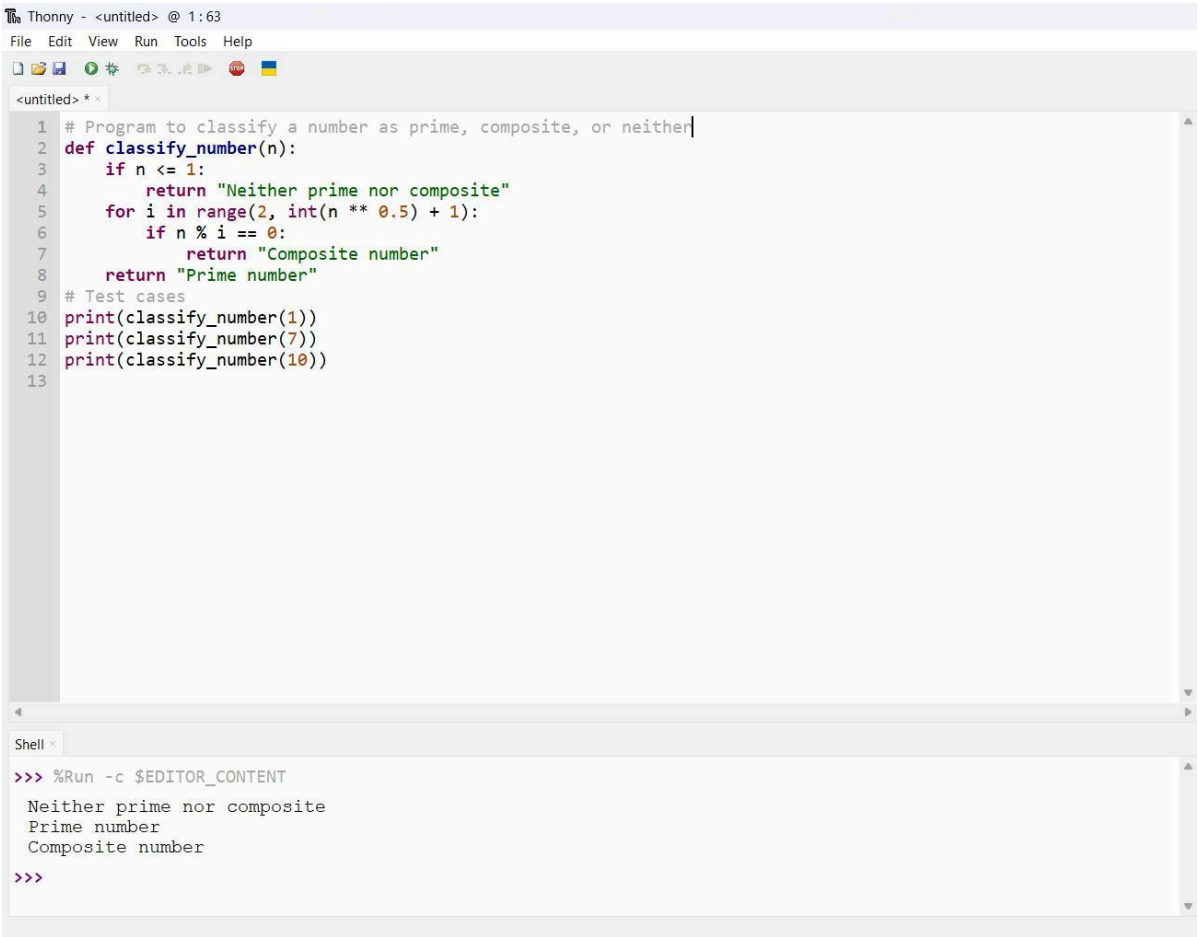
One-Shot Prompting – Factorial Calculation:-



Few-Shot Prompting – Armstrong Number Check:-



Question 4 (Optional): Context-Managed Prompting – Number Classification:-



Question 5: Zero-Shot Prompting – Perfect Number Check:-

Thonny - <untitled> @ 8 : 23

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```python
# Function to check whether a number is a perfect number
def is_perfect(num):
    if num <= 0:
        return False
    total = 0
    for i in range(1, num):
        if num % i == 0:
            total += i
    return total == num
# Test cases
print(is_perfect(6))
print(is_perfect(28))
print(is_perfect(12))
```

Shell

```
>>> %Run -c $EDITOR_CONTENT

 True
 True
 False

>>>
```

Few-Shot Prompting – Even or Odd with Validation:-

Thonny - <untitled> @ 8 : 21

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```python
# Program to check whether a number is even or odd with input validation
def even_or_odd(value):
    if not isinstance(value, int):
        return "Invalid input. Please enter an integer."
    if value % 2 == 0:
        return "Even"
    else:
        return "Odd"
# Test cases
print(even_or_odd(8))
print(even_or_odd(15))
print(even_or_odd(0))
print(even_or_odd(-7))
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Even
 Odd
 Even
 Odd
>>>
```